

Spenser Millburn – Resume/Portfolio

Robotics Software Engineer / Technologist

Qualifications

Dedicated, enthusiastic and ambitious professional with excellent comprehension and retention ability. Breadth and depth knowledge in a broad spectrum of technical fields concentrated within Mechatronics/Robotics, EE, Computer Science and Mechanical Design.

CS: C++, Python, Linux, JavaScript, React.js, Bash, *ROS(1,2)*, *OpenCV*, *Matplotlib*, *NumPy*, *QT (GUI)*, *Jupyter*, Control Systems, *Forward/Inverse Kinematics*, *Gazebo Simulation*, TCP/UDP, Websockets

EE: Advanced SMT Soldering, *Altium Designer*, I2C, SPI, CAN, Arduino/Raspberry Pi, FPGA, Oscilloscope, Logic Analyzer, Bed of Nails, Sensors & Actuators

ME: OnShape, Creo, Solidworks, AutoCAD, Additive MFG, CNC/CAM Machining (Mill/Lathe), DFM, Fluidics/Microfluidics, Thermal Analysis

Education

Northeastern University: Mechatronics Engineering, B.S.

Class of 2022, GPA 4.0

Central New Mexico Community College: Engineering, Associate of Science

Class of 2020, GPA: 4.0

Leadership – NEU Robotics, Quadruped Robotics Project Manager & Software Lead

- Active leadership of multidisciplinary team of 75+ engineering students working to develop advanced quadruped (Robot Dog) research platform.
- Highly involved with the architectural design of the system, established a cross functional software team and spearheaded 10+ subteams.
- Substantial technical contribution to the development of CAPSTAN servo reduction mechanisms, ROS (Robot Operating System), Simultaneous Localization and Mapping, Computer Vision, Inverse kinematics and more.

Practical Experience

Embedded Software Engineer – Alert Innovation & Walmart Global Tech

September 2022 – Current – Boston, MA

- Designed, implemented, and unit tested performant hierarchical state machines with modern C++17 in a real time QNX microkernel environment.
- Developed device drivers for critical I2C and SPI based sensors, enabling accurate localization of autonomous robots in production (50k units, 99.7% uptime).
- Spearheaded a company-wide transition towards Azure IoT based configuration management. Owned SDK integration to codebase development for embedded (C++), backend (Python), and frontend (React)
- Collaborated cross-functionally to establish data pipelines between IoT Hub and robot software, enabling efficient data exchange and real-time cloud analytics.
- Dockerized the embedded development environment, resulting in streamlined workflows, centralized tooling, and accelerated development cycles.

(March 2022- Current) – Robotics Engineer – Building Machines Incorporated

- Full ownership and architecture of a high powered (480V 50Hp) robotic fluid delivery system for additive manufacturing.

- Designed implemented tested and integrated a full stack React GUI to override physical control panel.
- WebSocket interface and Flask python backend. Systemd linux service management.
- IOT integration of multiple I2C and SPI sensors/Actuators with custom PCBs and C++ drivers.
- Modalities: Laser Time-Of-Flight, Depth CV, Pressure Transducers, Solenoid Valves, BLDC motor, Relays
- Data acquisition, analysis and visualization with JupyterLab and Python.
- Electromechanical design, build, test of fixtures

(March 2021- March 2022) – Engineering Technician – Owl Labs

- Rapid prototyping, prototype assembly and design. Supported 30+ FTEs.
- Professional fixture design, fabrication, testing and deployment. Contract manufacturing production line integration (10000+ units annual throughput)
- Thermal data acquisition and Test Engineering.
- Customer facing Embedded Linux and Android software development support.
- Data acquisition, analysis and visualization not limited to thermals, acoustics and airflow.

(August 2020- March 2021) – Engineering Technician – Degree Controls Systems

- Provide SMT rework and circuit analysis support.
- Project management of Engineering Change Order and Prototype Change Order documentation.
- Design PCBAs and electro-mechanical assemblies.
- Invented customer facing pythonic software, GUI design with PyQT5.
- Provide mechanical engineering CAD support of fixtures and jigs

(August 2018 – August 2020) – Operations Technician (Engineering Focused) – Sartorius NA

- Design, fabricate and document professional production jigs, ultimately generating a return greater than USD \$50,000 in automated labor and recovered inventory.
- Designed, prototyped and established custom IOT automation server framework, developed exclusively in python3
 - Automates paperwork system, specifically targeting key KPI and Metric calculations.
 - Utilizes custom microcontroller clients to report live production data via a socket server.
 - Automates label printing and generation.
 - Provides a dynamic project management platform, in order to achieve better insight and communication to managers via the server's web based GUI
- Successfully modeled and presented a restructured production floorplan to executive level administration
- One of three technicians to Manufacture, troubleshoot and certify highly accomplished equipment (specifically flow cytometry) (Unit price: \$330k USD)